

Project Report
Career Recommender
A Python-Based program for Student Career Guidance

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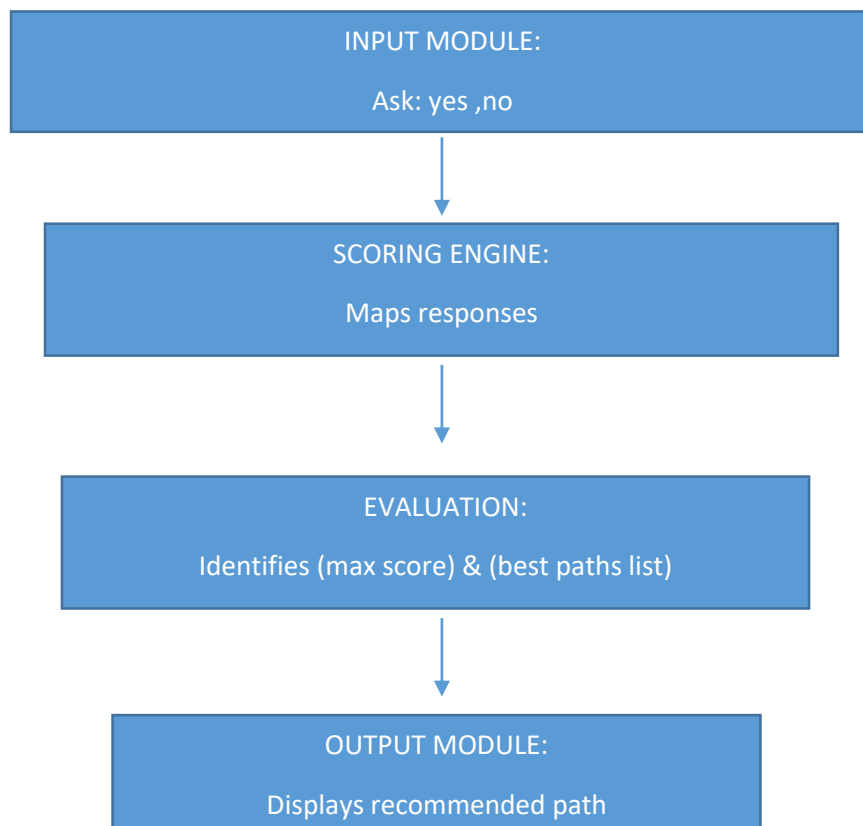
Introduction:

Students often face uncertainty about which technology specialization to pursue. This project develops a Python-based career interest recommender that helps students identify suitable tech domains through a simple questionnaire and rule-based scoring system.

Problem Statement:

Students often struggle with career direction, unsure which technology domain suits their interests—whether it's AI/ML, Web Development, Cybersecurity, Data Science, UI/UX Design, or Cloud & DevOps. This tool provides guidance to help students make decisions about their academic and career paths

System Architecture:



Implementation:

Sample Output:

===== Career Interest Recommender =====

Your Recommended Career Path is:

-> AI & Machine Learning

Interest level: High – This area strongly matches your interests!

What you should study first:

- Python basics and libraries like NumPy, Pandas
- Math: linear algebra, probability, statistics
- Intro to machine learning concepts

Future career options:

- Machine Learning Engineer
- Data Scientist
- AI Engineer

The system effectively differentiates between domains, handles ties with multiple recommendations, and provides actionable guidance.

Screenshots:

- **Code :**

C:\Users\ACER\OneDrive\Desktop\pythin learning term 1\project.py\careerrecommender.py

```
1 # Scores for each career path
2 scores = {
3     "AI & Machine Learning": 0,
4     "Web Development": 0,
5     "Cybersecurity": 0,
6     "Data Science": 0,
7     "UI/UX Design": 0,
8     "Cloud & Devops": 0}
9 # info: What Should You Study + Future options
10 study_plan = {
11     "AI & Machine Learning": {
12         "study": [
13             "Python basics and libraries like NumPy, Pandas",
14             "Math: Linear algebra, Probability, Statistics",
15             "Intro to machine learning concepts"],
16         "Future Occupation": [
17             "Machine Learning Engineer",
18             "Data Scientist",
19             "AI Engineer"]
20     },
21     "Web Development": {
22         "study": [
23             "HTML, CSS, basic JavaScript",
24             "Python with Flask or Django",
25             "Responsive web design basics"],
26         "Future Occupation": [
27             "Frontend Developer",
28             "Backend Developer",
29             "Full Stack Web Developer"]
30     },
31     "Cybersecurity": {
32         "study": [
33             "Computer networks and OS basics",
34             "Linux commands and system security basics",
35             "Common attacks (phishing, malware, SQL injection)"],
36         "Future Occupation": [
37             "Cybersecurity Analyst",
38             "Ethical Hacker"]
39     },
40     "Data Science": {
41         "study": [
42             "Statistics and Probability",
43             "Machine Learning",
44             "Data Visualization"],
45         "Future Occupation": [
46             "Data Scientist",
47             "Data Analyst",
48             "Business Intelligence Analyst"]
49     },
50     "UI/UX Design": {
51         "study": [
52             "User-centered design",
53             "Wireframing and Prototyping",
54             "Visual Design Principles"],
55         "Future Occupation": [
56             "UX Designer",
57             "UI Designer",
58             "Product Designer"]
59     },
60     "Cloud & Devops": {
61         "study": [
62             "Cloud Fundamentals (AWS/Azure/GCP)",
63             "Containerization (Docker, Kubernetes)",
64             "Infrastructure as Code (Terraform)"],
65         "Future Occupation": [
66             "Cloud Architect",
67             "DevOps Engineer",
68             "Site Reliability Engineer"]
69     }
70 }
```

```

42         "Security Engineer"]
43     },
44
45     "Data Science": {
46         "study": [
47             "Python, statistics, and data analysis",
48             "Pandas, NumPy, and data visualization",
49             "Basic machine learning algorithms"],
50         "Future Occupation": [
51             "Data Analyst",
52             "Data Scientist",
53             "Business Intelligence Engineer"]
54     },
55
56     "UI/UX Design": {
57         "study": [
58             "Design principles and color theory",
59             "Wire framing tools (Figma, Adobe XD basics)",
60             "User research and usability testing"],
61         "Future Occupation": [
62             "UI Designer",
63             "UX Designer",
64             "Product Designer"]
65     },
66
67     "Cloud & Devops": {
68         "study": [
69             "Basics of Linux and command line",
70             "Intro to cloud platforms (AWS, Azure, GCP)",
71             "Version control (Git) and CI/CD basics"],
72         "Future Occupation": [
73             "Cloud Engineer",
74             "Devops Engineer",
75             "Site Reliability Engineer"]
76     }
77 }
78

```

```

78 }
79
80 print("----- Career Interest Recommender -----")
81 print()
82 print("Answer the following questions with: 1 (Yes) or 0 (No)")
83 print()
84
85 # Helper: get 0/1 safely
86 def ask_yes_no(question):
87     while True:
88         ans = input(question + " ")
89         if ans in ["0", "1"]:
90             return int(ans)
91         print("Please enter only 1 for Yes or 0 for No.")
92
93 # 6 short questions
94 q1 = ask_yes_no("Do you enjoy solving logical or math problems? (1/0)")
95 if q1==1:
96     scores["AI & Machine Learning"] += 2
97     scores["Data Science"] += 2
98     scores["Cybersecurity"] += 1
99
100
101 q2 = ask_yes_no("Do you like creating or designing visually appealing things? (1/0)")
102 if q2==1:
103     scores["UI/UX Design"] += 2
104     scores["Web Development"] += 1
105
106
107 q3 = ask_yes_no("Are you curious about how websites and apps are built? (1/0)")
108 if q3==1:
109     scores["Web Development"] += 2
110     scores["Cloud & Devops"] += 1
111
112
113 q4 = ask_yes_no("Do you like finding patterns in data or charts? (1/0)")
114 if q4==1:
115     scores["Data Science"] += 2
116     scores["AI & Machine Learning"] += 1

```

```

117
118
119 q5 = ask_yes_no("Are you interested in protecting systems from hackers? (1/0)")
120 if q5==1:
121     scores["Cybersecurity"] += 2
122     scores["Cloud & Devops"] += 1
123
124
125 q6 = ask_yes_no("Do you enjoy working with tools, automation, or cloud platforms? (1/0)")
126 if q6==1:
127     scores["Cloud & Devops"] += 2
128     scores["AI & Machine Learning"] += 1
129
130
131 # Find best paths
132 max_score = max(scores.values())
133 best_paths = [name for name, val in scores.items() if val==max_score]
134
135
136 # Interest level
137 if max_score <= 2:
138     interest_level = "Low - You are still exploring different areas."
139 elif max_score <= 4:
140     interest_level = "Medium - This area matches some of your interests."
141 else:
142     interest_level = "High - This area strongly matches your interests!"
143
144 print("\n===== Result =====\n")
145
146 if len(best_paths)==1:
147     print("Your Recommended Career Path is:")
148 else:
149     print("You match well with multiple career paths:")
150
151 for path in best_paths:
152     print("->", path)
153
154 print("\nInterest level:",interest_level)
155
156 print("\nScores for reference:")
157 for name,val in scores.items():
158     print(name,":",val)
159
160 # Show what to study and future options for each best path
161 print("\n===== What You Should Study & Future Options =====\n")
162
163 for path in best_paths:
164     info = study_plan[path]
165     print("Career Path:", path)
166     print("What you should study first:")
167     for item in info["study"]:
168         print(" -", item)
169     print("Future career options:")
170     for job in info["Future Occupation"]:
171         print(" -", job)
172     print()
173
174 print("Note:This is only a simple guide.")

```

- Output:

```

PS C:\Users\ACER\OneDrive\Desktop\python learning term 1> cd C:\Users\ACER\AppData\Local\Programs\Python\Python39-64\Scripts
----- Career Interest Recommender -----

Answer the following questions with: 1 (Yes) or 0 (No)

Do you enjoy solving logical or math problems? (1/0) 1
Do you like creating or designing visually appealing things? (1/0) 0
Are you curious about how websites and apps are built? (1/0) 1
Do you like finding patterns in data or charts? (1/0) 1
Are you interested in protecting systems from hackers? (1/0) 1
Do you enjoy working with tools, automation, or cloud platforms? (1/0) 0

===== Result =====

Your Recommended Career Path is:
-> Data Science

Interest level: Medium - This area matches some of your interests.

Scores for reference:
AI & Machine Learning : 3
Web Development : 2
Cybersecurity : 3
Data Science : 4
UI/UX Design : 0
Cloud & Devops : 2

===== What You Should Study & Future Options =====

Career Path: Data Science
What you should study first:
- Python, statistics, and data analysis
- Pandas, NumPy, and data visualization
- Basic machine learning algorithms
Future career options:
- Data Analyst
- Data Scientist
- Business Intelligence Engineer

Note: This is only a simple guide.
PS C:\Users\ACER\OneDrive\Desktop\python learning term 1> █

```

Learnings & Key Takeaways:

- **Data Structure:** Use of dictionary, nested if-else blocks, functions, loop.

Future Enhancements:-

Potential improvements for advanced versions:

- Database Integration: Store user responses and results for analytics
- Advanced Scoring: Implement weighted algorithms or machine learning
- Graphical Interface: Create a GUI version using tkinter or PyQt
- Skill Assessment: Add follow-up questions to measure existing knowledge
- Export Results: Save results as PDF or Excel files
- Personalized Roadmaps: Generate detailed learning timelines

References:

- Python programming using problem solving approach - reema thareja
- Programming and problem solving with python - ashok kamthame, amit kamthame
- Let us python - yashavant kanetkar, Aditya kanetkar